THE FUNCTOR C SENDS FLAT MODULES TO INJECTIVE MODULES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted spans carry 2 navigation stubs, for the adjunction identity and for faithfulness of C; the three rewriting steps themselves are displayed in the span.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Editorial note. The canonical tag of this label does not occur among the tags of the published items of descent.tex, so it is not a published duplicate.

Allowed standard prerequisites (editorial).

- The adjunction identity rewriting Hom into C as C of a tensor product. Same chapter, equation-adjunction, invoked at proof lines 1565-1569. Supplies the rewriting of the second sequence into the Hom sequence.
- Faithfulness of the functor C. Same chapter, Lemma lemma-C-is-faithful, invoked at proof lines 1561-1565. Supplies exactness of the sequence after applying C.
- Flatness as exactness of tensoring with M. Standard module theory, used at proof lines 1555-1560. Supplies exactness of the tensor product of the test sequence with M.

Source reference. Original equation 08WT: sources/stacks-descent/pinned-original-source.tex, source lines 1195–1195. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 08WR: sources/stacks-descent/pinned-original-source.tex, source lines 1179–1179. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *If $M \in Mod_R$ is flat, then $C(M)$ is an injective R-module.*

Proof. Let $0 \to N \to P \to Q \to 0$ be an exact sequence in Mod_R. Since M is flat,

$$0 \to N \otimes_R M \to P \otimes_R M \to Q \otimes_R M \to 0$$

is exact. By Lemma 08WR,

$$0 \to C(Q \otimes_R M) \to C(P \otimes_R M) \to C(N \otimes_R M) \to 0$$

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is exact. By (08WT), this last sequence can be rewritten as

$$0 \to \mathrm{Hom}_R(Q, C(M)) \to \mathrm{Hom}_R(P, C(M)) \to \mathrm{Hom}_R(N, C(M)) \to 0.$$

Hence $C(M)$ is an injective object of Mod_R. $\square$